

Trainer Debrief — Lab 10

Sampling Strategy, Sample Size and Rational Subgrouping

Case study: Meridian Medical Devices — seal weld burst pressure

Objective: Determine defensible sample sizes and design rational subgroups for your capstone (A4).

***TRAINER ONLY.** Do not hand this out before the lab is complete.*

Expected analysis

lab-10-sampling-frame.xlsx

The population mean is ~214 kPa with SD ~21.5. A simple random sample of n=30 estimates the mean well but hides the plant-to-plant difference; a sample stratified by plant exposes it. For a margin of error of 5 kPa at 95% confidence, $n = (1.96 \times 21.5 / 5)^2 \sim 71$ units.

Every dataset is generated from a fixed seed, so these figures are reproducible — re-run the analysis on the shipped workbook and you will get these numbers.

What good looks like

Your sample sizes are calculated from a real estimate of s or p, your subgrouping scheme places common-cause variation within subgroups, and you can state the shift size your plan can detect.

Where this sits in the capstone

Re-derive sampling schemes and sample size formulas in full, then go beyond Green Belt with RATIONAL SUBGROUPING — the decision that silently determines what your control charts in Day 4 will be capable of detecting at all. CAPSTONE BUILDING BLOCK — the output of this lab goes directly into your capstone project pack, which you present to the steering committee on Day 5.

The running process — figures that must reconcile

Checkpoint	Value	Established in
Baseline defect rate	2.9%	Lab 3
Measurement system	FAILS Gage R&R (~40% study variation)	Lab 11
Baseline capability	Pp 0.62 / Ppk 0.53	Lab 14
Proven Xs	clamp pressure, dwell time, electrode tip age	Labs 18-21
DOE optimum	clamp 47.4 bar, dwell 2.76 s	Labs 23-25
Improved capability	Ppk 1.40	Lab 27
Annual hard benefit	SGD 487,000 vs SGD 812,000 COPQ	Labs 1, 30

If a learner's number disagrees with the column above, they have either used their own workplace data (fine — say so) or made an error (find it).

